engine. Such is the versatility of this material that a nozzle of 1 mm thickness withstands the great pressure and temperature as the rocket fires away for several minutes. HS-25 also offered great workability, which means it can be made into peculiar shapes without inflicting stress points on the material. This made HS-25 ideal for the bell-shaped nozzle used by the Viking engine at high altitude. While a conical nozzle can be made in a conventional lathe using other alloys, it takes a versatile material like HS-25 to make into a bell shape by using a pressure mechanism to flow it.

After scanning details of the best among Indian centres that can develop high-end hardware, I zeroed in on Mishra Dhatu Nigam Limited (MIDHANI). As a government unit with ample facilities to give it a shot, MIDHANI fitted our bill, but the bureaucracy there was stifling. With the inherent inertia of any Indian public sector unit, the MIDHANI top brass just refused to play ball.

The other option was ISRO's own metallurgy team headed by M K Mukherjee and Sarkar as his deputy. Sarkar started working on finding an equivalent for a material called Shell-407 for control rockets.

Shell-407 was a brand supplied by the American company Shell, and the real material content remained undisclosed to ISRO. When injected with hydrozine, this material decomposes, releasing a gas that gives thrust for course correction of the spacecraft. In another example of how the US systematically stymied ISRO's efforts in rocketry, just when the metallurgy team was experimenting with Shell-407, the company, which was providing the same material to NASA, kept on increasing the price for India and, finally, stopped supplying it.

They said it had been classified as strategic material. Any scientist of the times knew that only those materials used in missiles and nuclear devices were categorised as 'strategic', and nobody – the Americans included – was using Shell-407 for missiles or nuclear devices.

This was the same tactics the US had employed to stop the USSR from transferring the cryogenic engine technology to India in the early 1990s. The US argued that cryogenic engines fell under technology for missiles. This was far from the truth. No country, including the US, has a missile with a cryogenic engine.

But already under the strain of the impending fall of the Soviet Union, the Russian facilities buckled under American pressure and changed the contract to